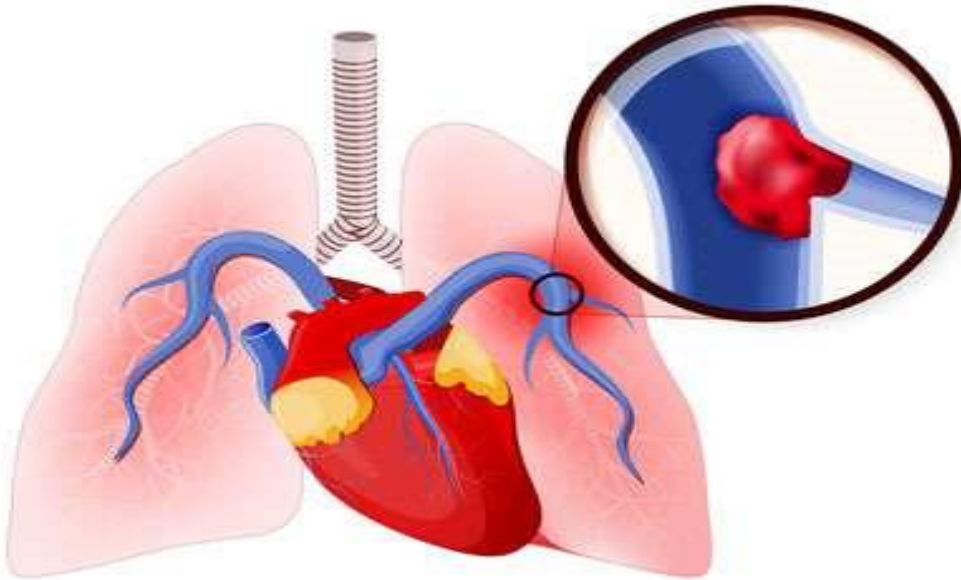
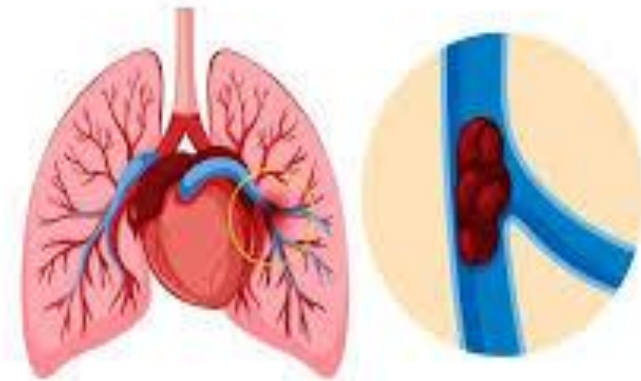


Pulmonary Embolism (PE)

Pulmonary embolism



Pulmonary Embolism (PE)



Prepared by

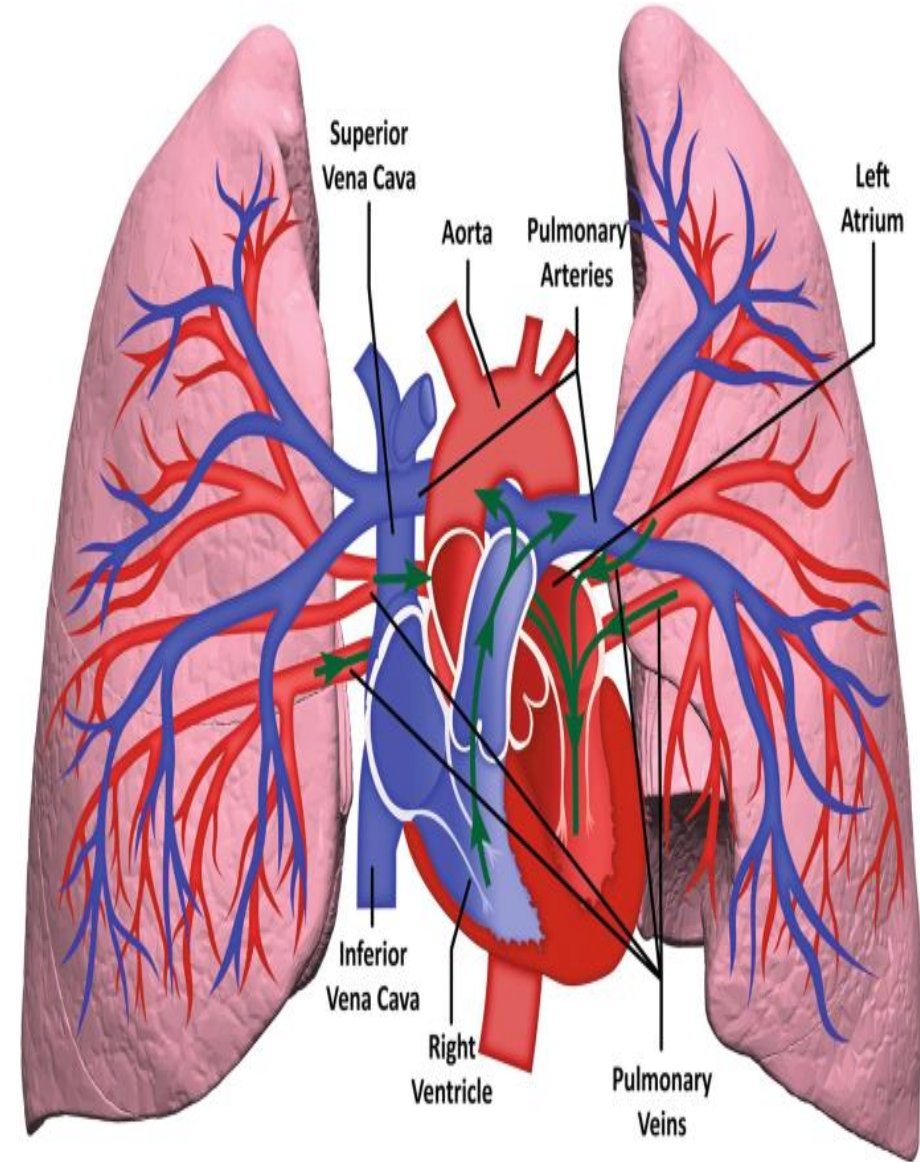
Assist professor Dr/Afaf Mohamed Mansour

Outlines

- ❖ Anatomy of pulmonary arteries
- ❖ Description
- ❖ Etiology
- ❖ Pathophysiology
- ❖ Assessment and Diagnosis
- ❖ Medical Management
- ❖ Nursing Intervention

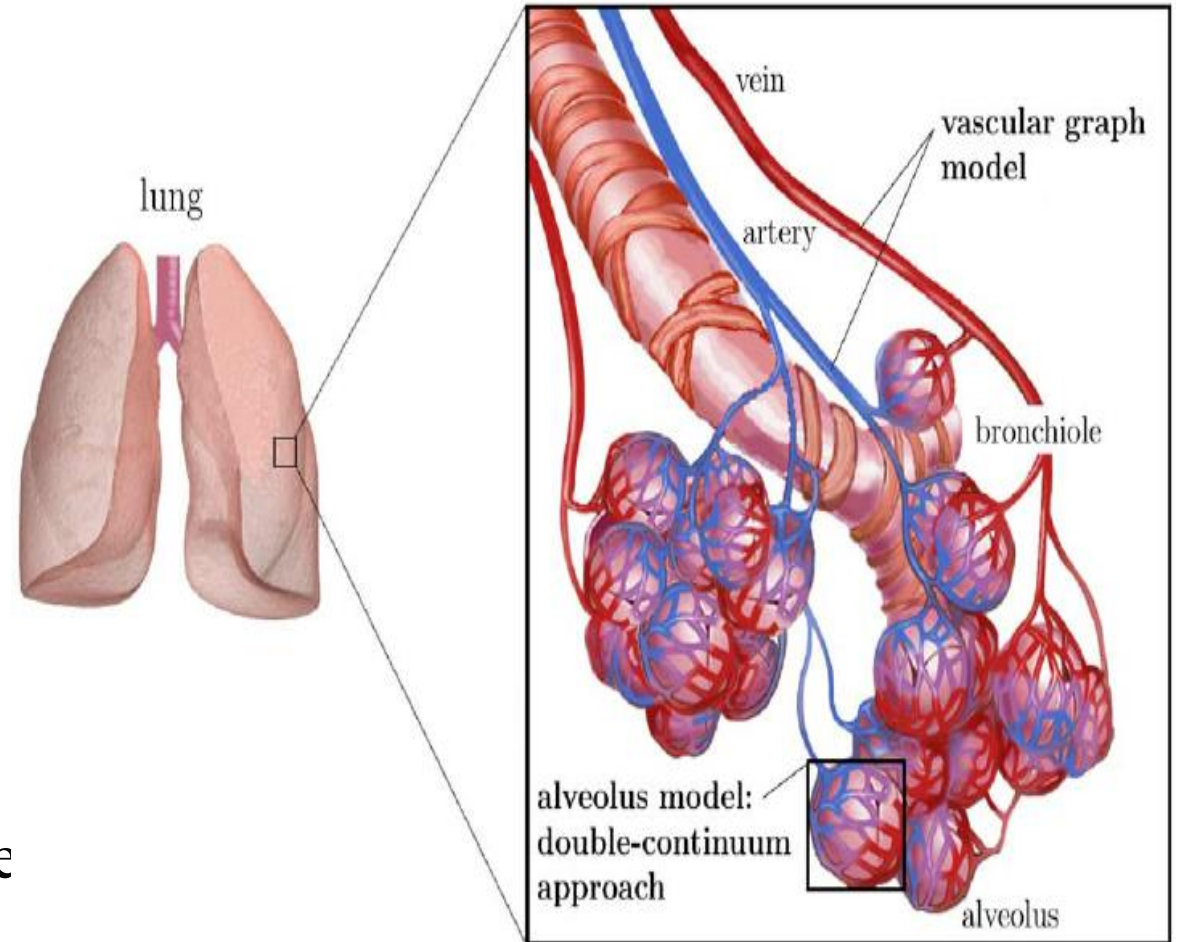
Anatomy of pulmonary arteries

- **Main pulmonary artery (Pulmonary trunk):** A short, wide vessel that originates from the right ventricle of the heart. It is one of the "great vessels" of the heart, along with the aorta.
- **Right and left pulmonary arteries:** The main pulmonary artery divides into these two branches beneath the aortic arch. The right pulmonary artery carries blood to the right lung, and the left pulmonary artery carries blood to the left lung.



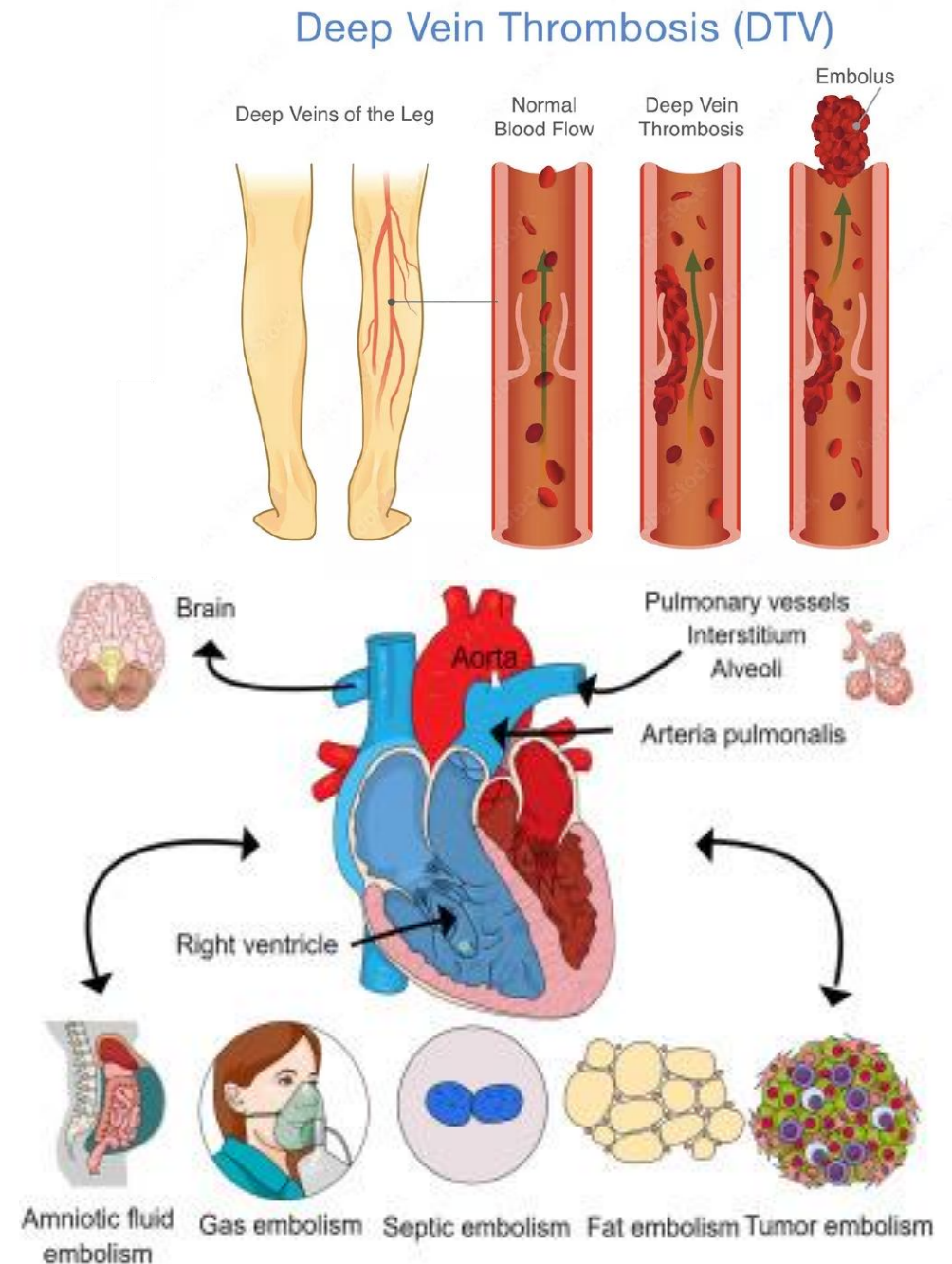
Anatomy of pulmonary arteries

- **Smaller branches:** Upon entering the lungs, the main arteries split further into lobar arteries, then into segmental, subsegmental, and finally, into tiny arterioles.
- **Capillary network:** The arterioles terminate in capillary beds that surround the alveoli, which is where gas exchange takes place.



Description

A pulmonary embolism (PE) occurs when a clot (thrombotic embolus) or other matter (non thrombotic embolus) lodges in the pulmonary arterial system, disrupting the blood flow to the lungs. **The majority of thrombotic emboli** arise from the deep leg veins, particularly the **iliac, femoral, and popliteal veins** . Other sources include the right ventricle, the upper extremities, and the pelvic veins. **Non-thrombotic emboli** arise from **fat, tumors, amniotic fluid, air, and foreign bodies**.

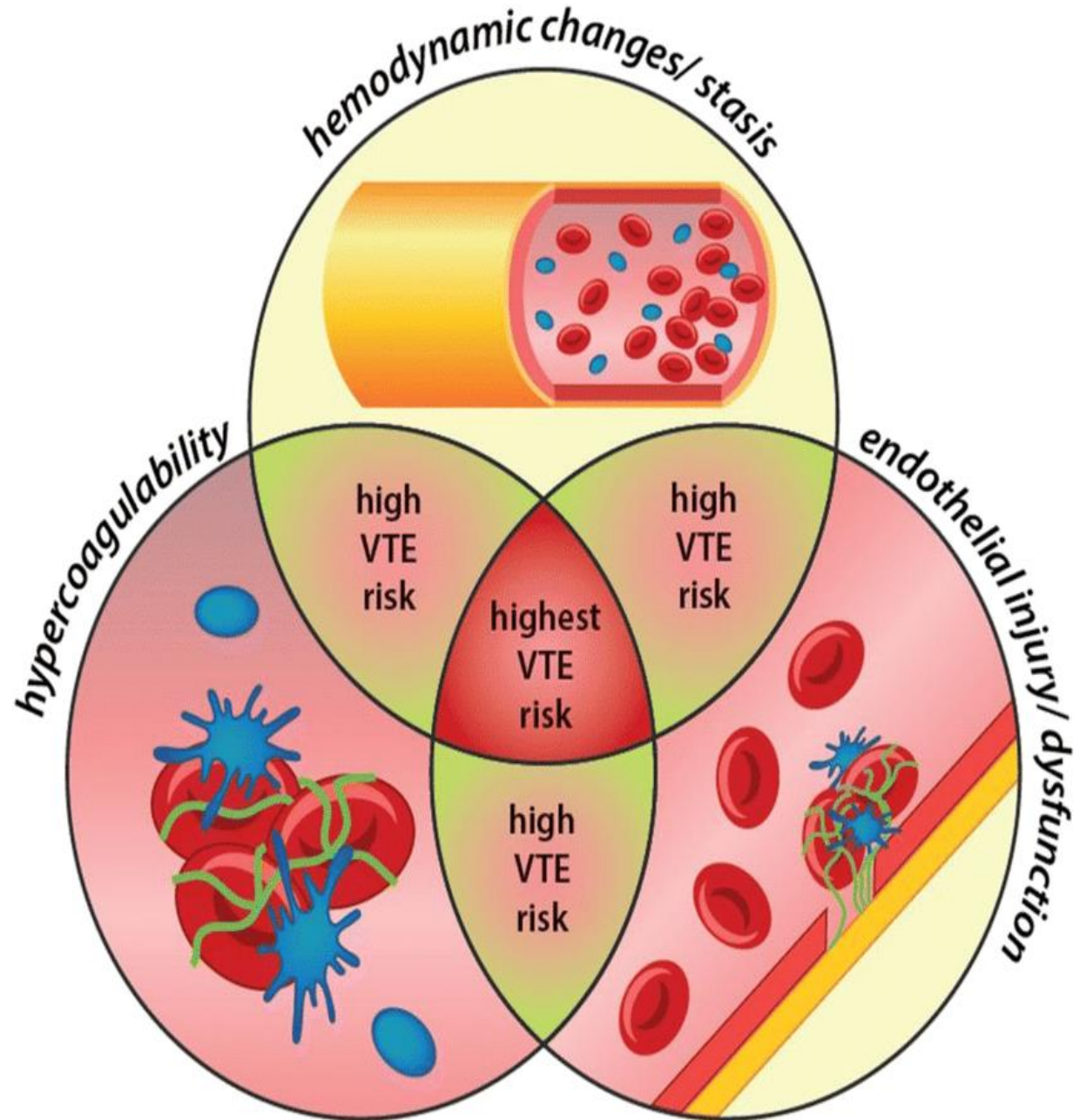


Etiology

A number of predisposing factors and precipitating conditions put a patient at risk for developing a pulmonary embolism.

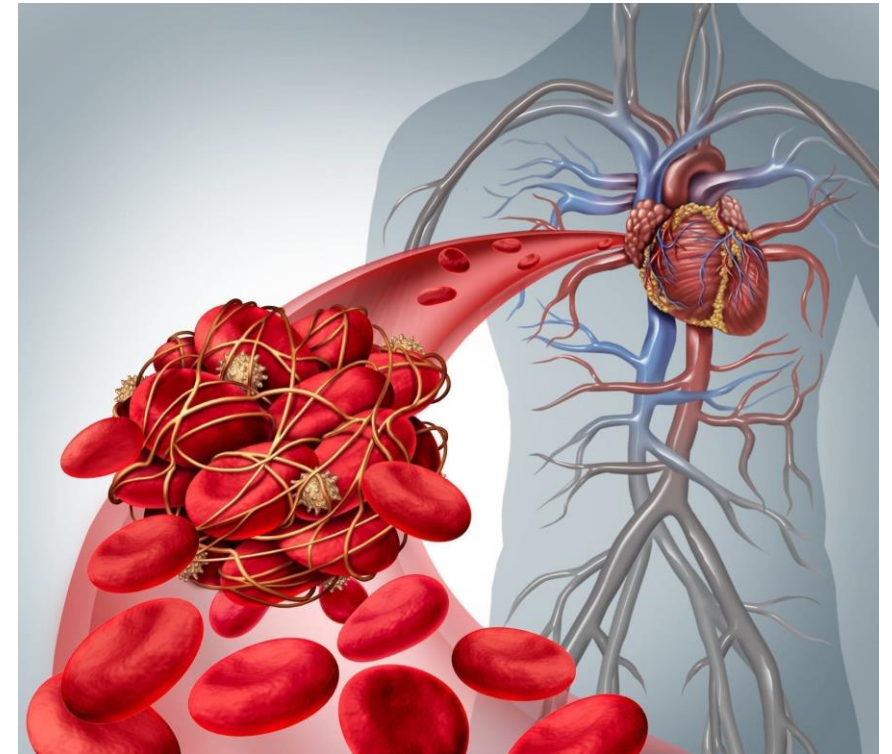
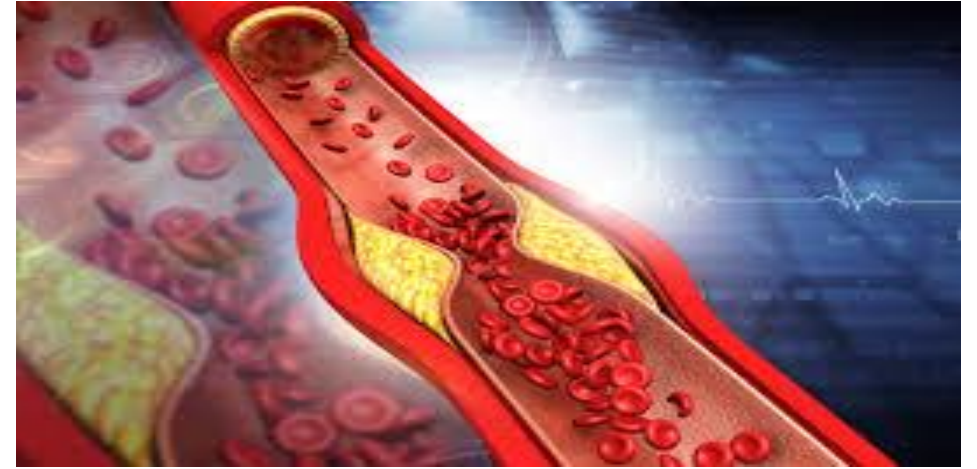
Three main predisposing factors

- ☐ Hypercoagulability,
- ☐ Injury to vascular endothelium
- ☐ Venous stasis (Virchow's triad)



• Causes of the venous stasis.

- Atrial fibrillation
- Immobility
- Injury to vascular endothelium
- Local vessel injury
- Atherosclerosis
- Hypercoagulability
- Polycythemia



Precipitating Conditions

- Previous pulmonary embolus
- Cardiovascular disease
 - Heart failure
 - Cardiomyopathy
 - Cor pulmonale

- **Surgery**

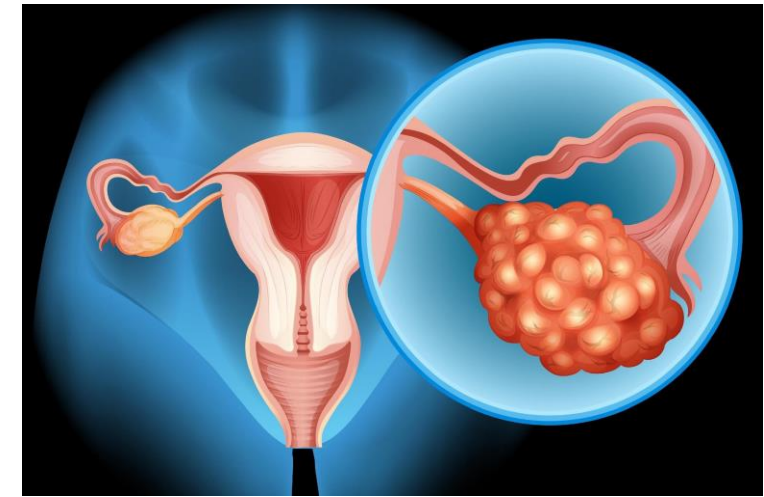
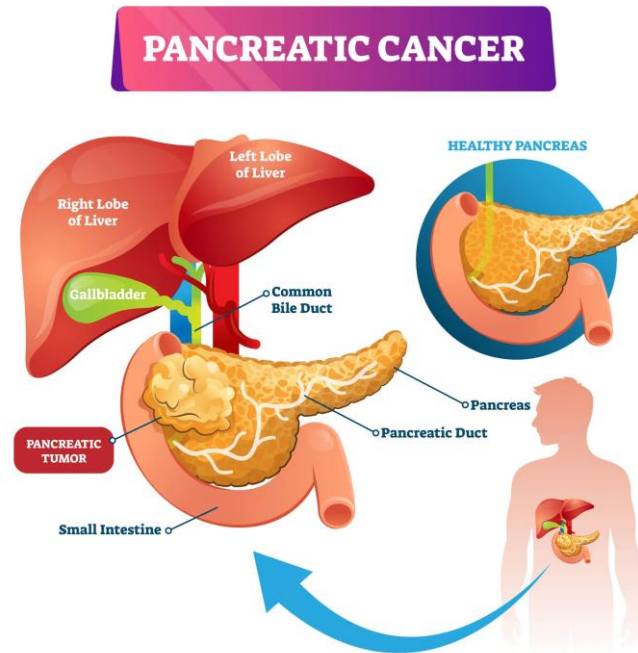
- Orthopedic
- Vascular
- Abdominal



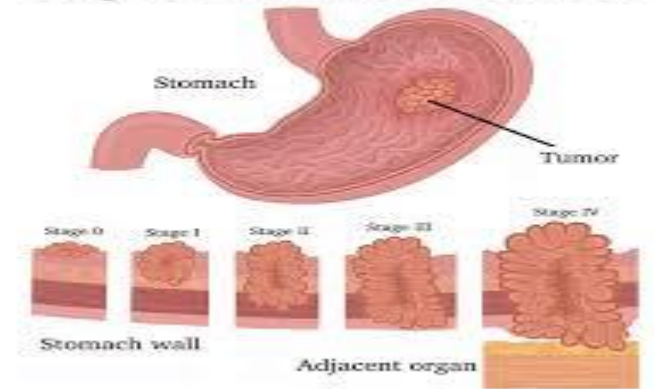
Precipitating Conditions

Cancer

- ❖ Ovarian
- ❖ Pancreatic
- ❖ Stomach

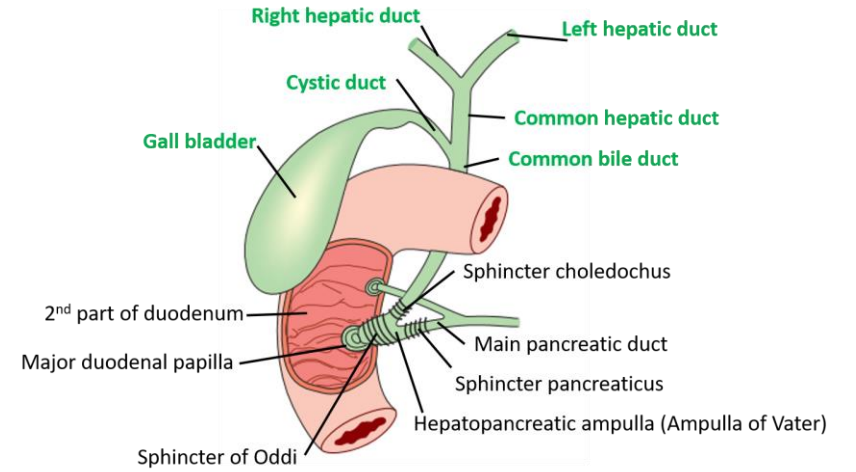


Stages of stomach cancer



Precipitating Conditions

- **Extrahepatic bile duct system**
- **Trauma (injury or burns)**
 - Lower extremities
 - Pelvis
 - Hip
- **Gynecologic status**
- Pregnancy
- Postpartum
- Birth control pills



Pathophysiology

A massive PE occurs with the blockage of a lobar or larger artery, resulting in **occlusion of more than 40% of the pulmonary vascular bed.**

Blockage of the pulmonary arterial system has both **pulmonary** and **hemodynamic** consequences.

The effects on the pulmonary system are

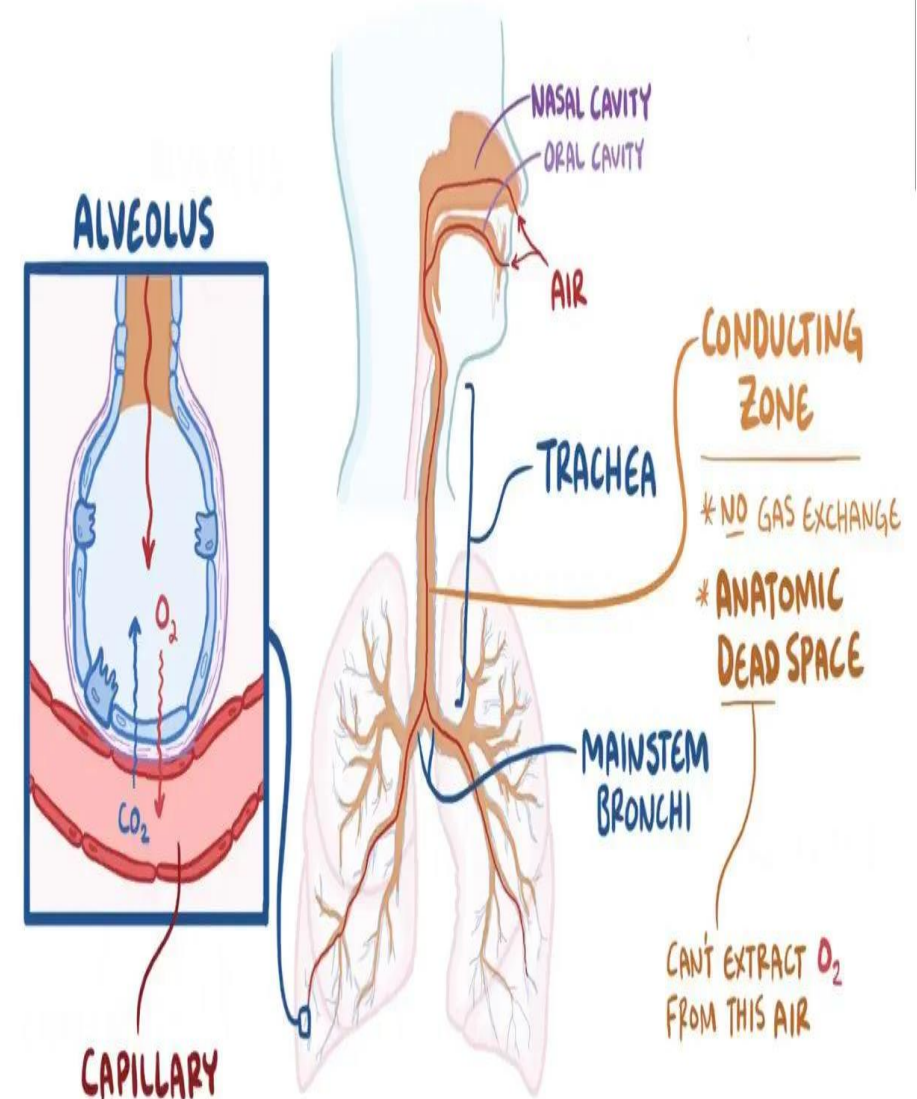
- ☐ Increased alveolar dead space
- ☐ Bronchoconstriction,
- ☐ Compensatory shunting.

☐ **The hemodynamic effects include**

- ☐ An increase in pulmonary vascular resistance
- ☐ Right ventricular workload

Increased Dead Space

1. An increase in alveolar dead space occurs because an area of the lung is receiving ventilation without being perfused .
2. The ventilation to this area is known as wasted ventilation, because it does not participate in gas exchange.
3. This effect leads to **alveolar dead space ventilation** and **an increase in the work of breathing** to limit the amount of dead space ventilation, localized bronchoconstriction occurs.



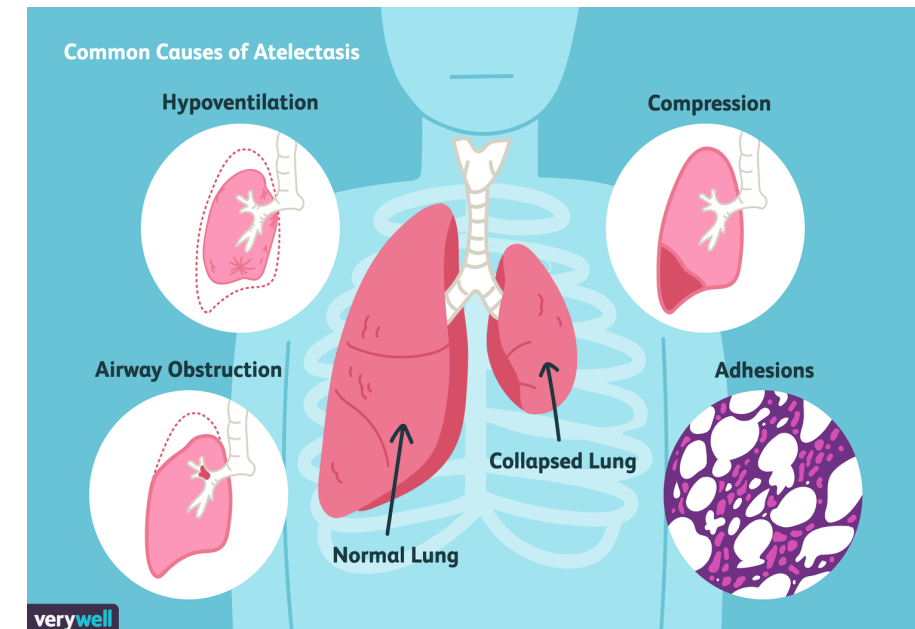
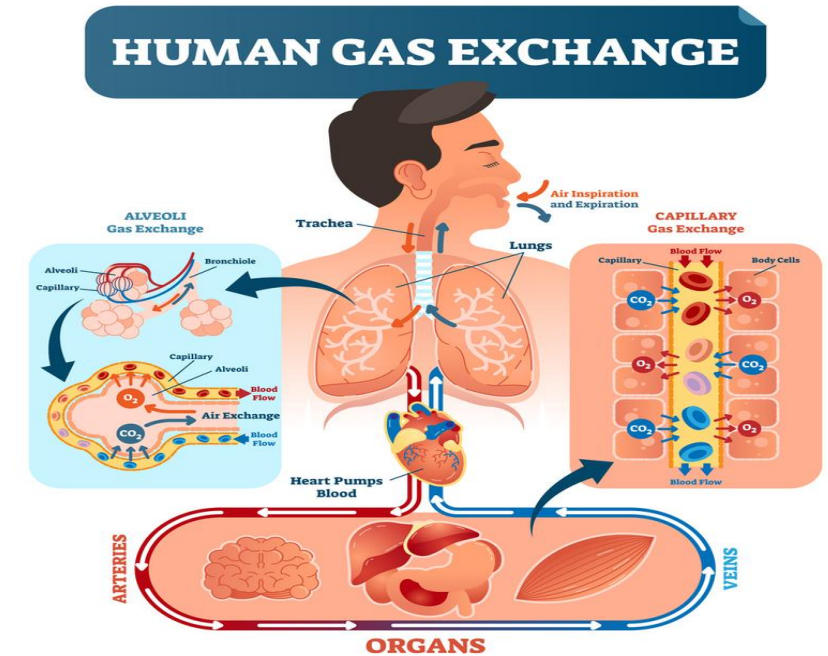
Bronchoconstriction

Bronchoconstriction develops as a result of **alveolar hypocarbia, hypoxia, and the release of mediators.**

1. Alveolar hypocarbia occurs as a consequence of

- Decreased carbon dioxide in the affected area and leads to constriction of the local airways
- Increased airway resistance
- Redistribution of ventilation to perfused areas of the lungs.

2- A variety of mediators are released from the site of the injury, either from the clot or the surrounding lung tissue, which further causes constriction of the airways. Bronchoconstriction promotes the development of **atelectasis.**



Compensatory Shunting

1. Compensatory shunting occurs as a result of the unaffected areas of the lungs having to accommodate the **entire Cardiac Output**.
2. This creates a situation in which perfusion exceeds ventilation and blood is returned to the left side of the heart without participating in gas exchange. This leads to the development of **hypoxemia**

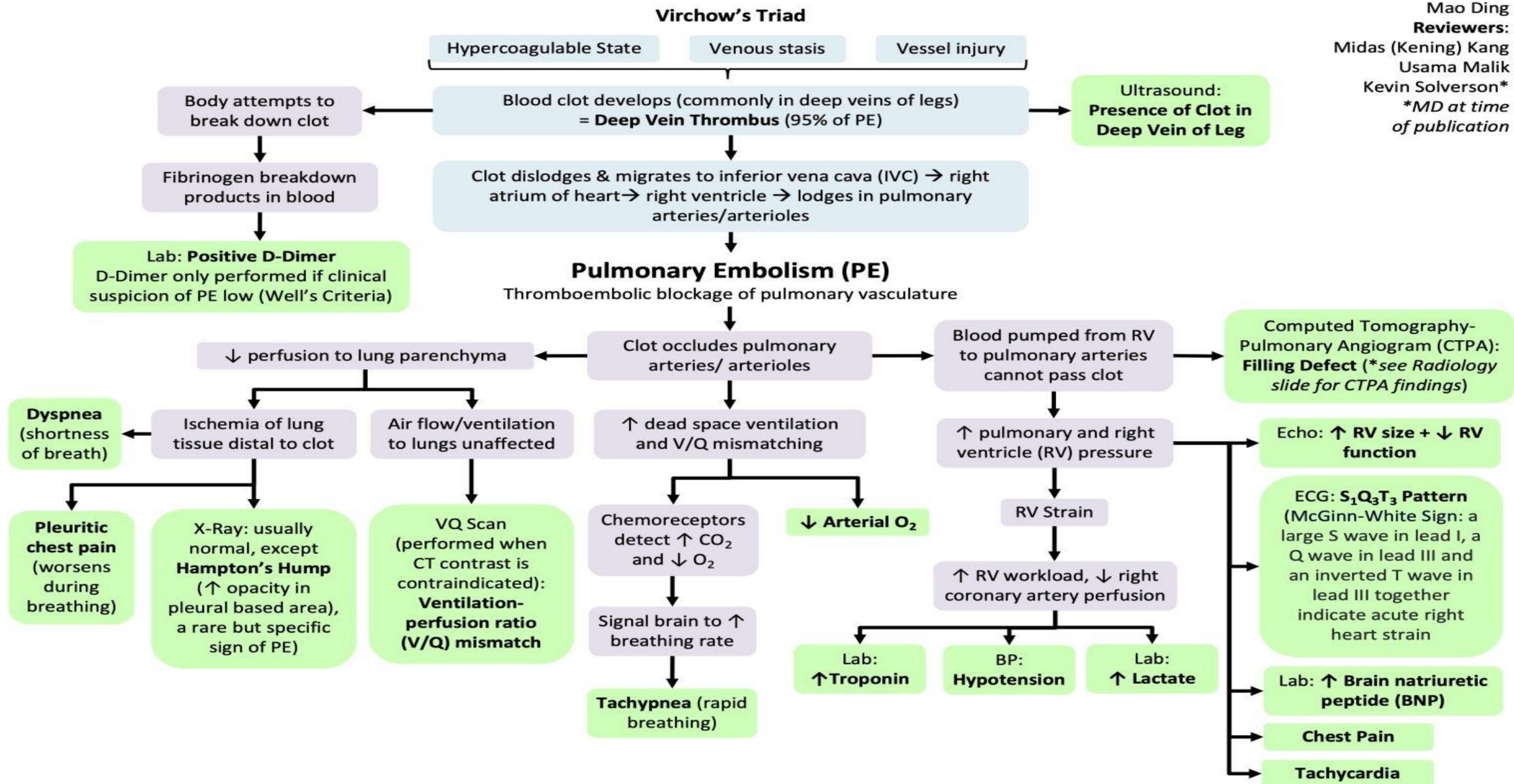
Hemodynamic Consequences

The major hemodynamic consequence of a PE is the development of pulmonary hypertension, which is part of the effect of a mechanical obstruction when more than 50% of the vascular bed is occluded.

- In addition, the **mediators released** at the injury site and the development of **hypoxia cause pulmonary vasoconstriction**, which further exacerbates pulmonary hypertension.
- As the **pulmonary vascular resistance increases**, so does the workload of the right ventricle as reflected by a rise in pulmonary artery (PA) pressures.
- Consequently, right ventricular failure occurs, which can lead to decreases in left ventricular preload, CO, and blood pressure, and shock.

Pulmonary Embolism: *Pathogenesis and Clinical Findings*

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Assessment and Diagnosis

The patient with a PE may have any number of presenting signs and symptoms

- Tachycardia and tachypnea
- Dyspnea
- Apprehension
- Fever
- Crackles
- Pleuritic chest pain
- Cough
- Hemoptysis
- Syncope and hemodynamic instability can occur as a result of right ventricular failure.

Initial laboratory studies and diagnostic procedures:

- ABG analysis
- D-dimer
 - Fibrinogen breakdown in the blood
- Electrocardiogram (EGG)
- Chest radiography, and Echocardiography (ECHO)

ABGs may show

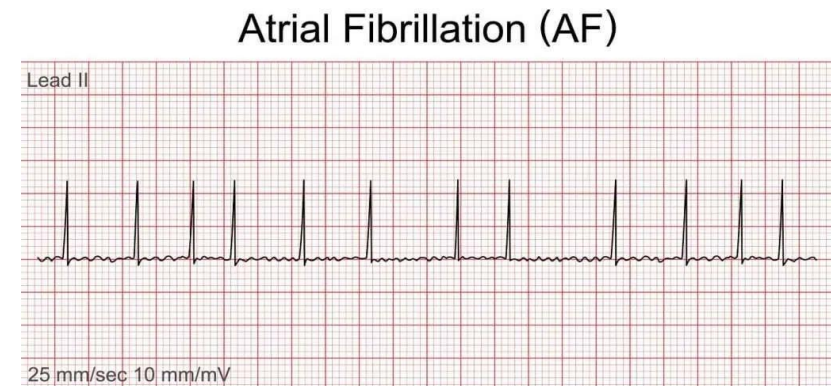
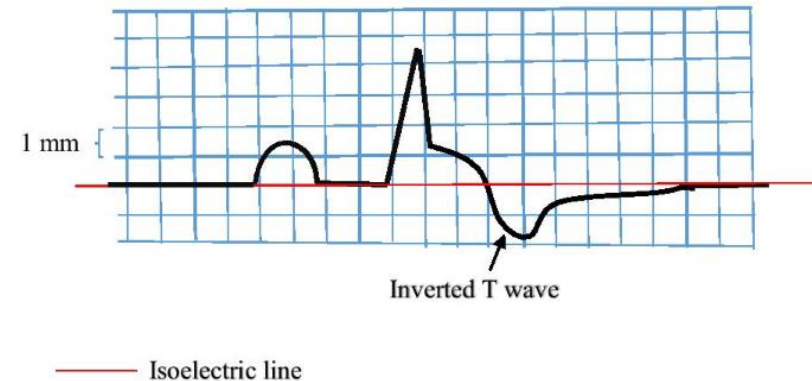
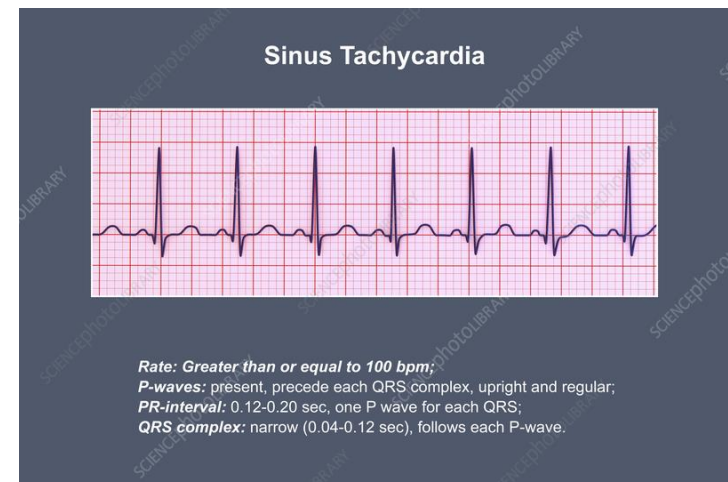
- **A low Pao₂**, indicating hypoxemia
- **A low Paco₂**, indicating hypocarbia
- **A high pH**, indicating **a respiratory alkalosis**
- The hypocarbia with resulting respiratory alkalosis is caused by tachypnea.

ECG interpretation PE

The most frequent ECG finding seen in the patient with a PE is **sinus tachycardia**.

The classic ECG pattern associated with a PE :**inverted T wave**

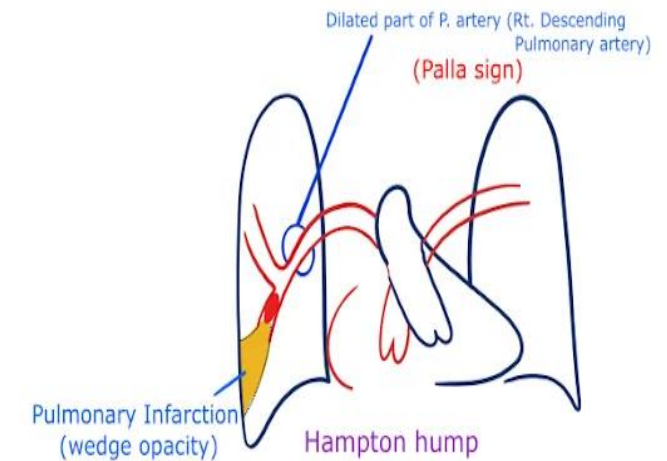
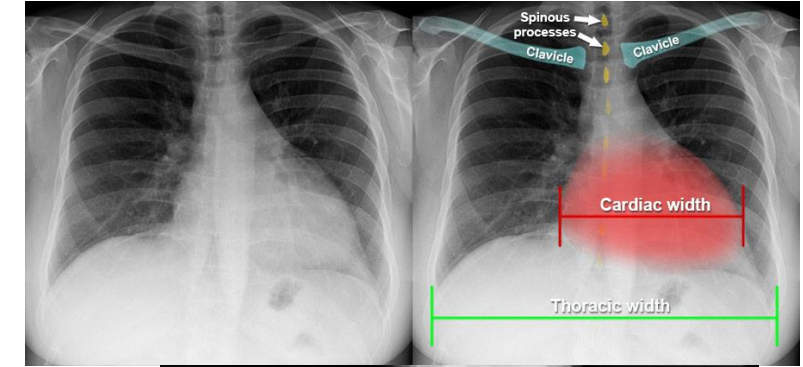
Other ECG findings associated with a PE include **right bundle branch block**, **new-onset atrial fibrillation**.



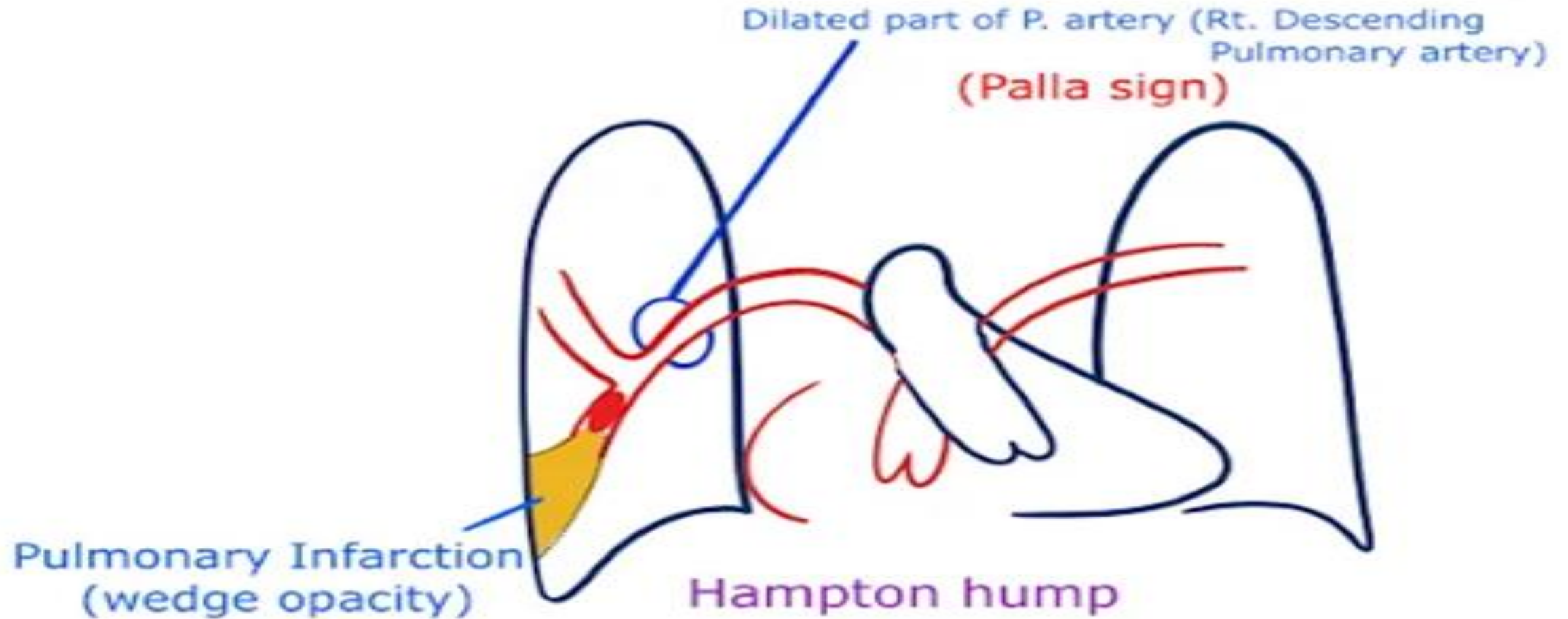
Assessment and Diagnosis

-**Chest x-Ray findings** Abnormal findings include

- ❖ Cardiomegaly
- ❖ Pleural effusion
- ❖ Elevated hemidiaphragm
- ❖ Enlargement of the right descending pulmonary artery (Palla's sign) a wedge-shaped density above the diaphragm (Hampton's hump) and the presence of atelectasis.

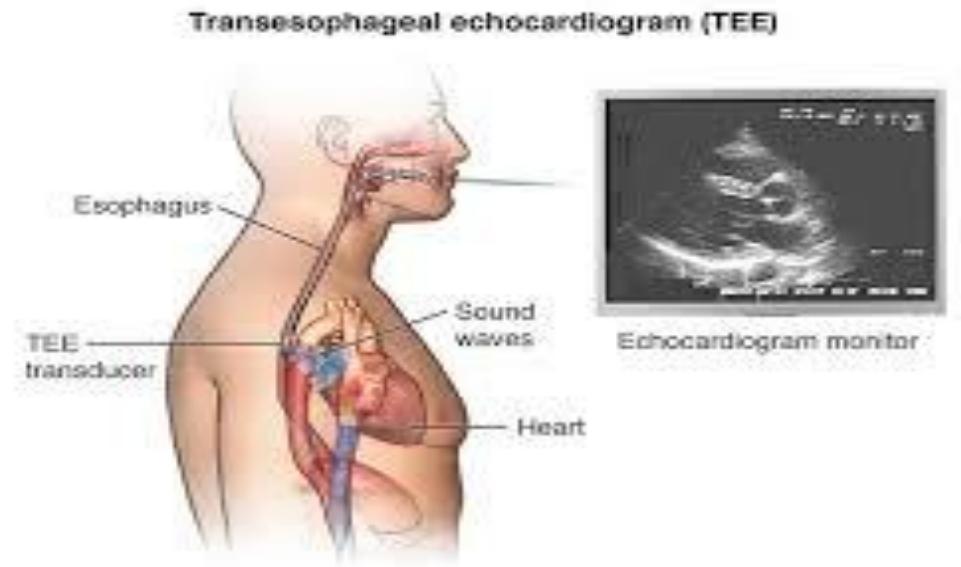


(Palla's sign) a wedge-shaped



Assessment and Diagnosis

An **ECHO**, either transthoracic or transesophageal, is also useful in the identification of a PE, because it can provide **visualization of any emboli in the central pulmonary arteries**



Medical Management

Medical management of the patient with PE

involves

- **Prevention strategies.**
- **Treatment strategies.**

Medical Management

Preventive strategies include:

- ❖ Use of prophylactic anticoagulation with low-dose or adjusted-dose heparin, LMWH, or oral anticoagulants.
- ❖ Use of pneumatic compression has also been demonstrated as effective methods of prophylaxis in low-risk patients.



Medical Management

Treatment strategies Include

- ☐ Preventing the recurrence of PE
- ☐ Facilitating clot dissolution
- ☐ Reversing the effects of pulmonary hypertension
- ☐ Promoting gas exchange
- ☐ Preventing complications.

Prevention of Recurrence

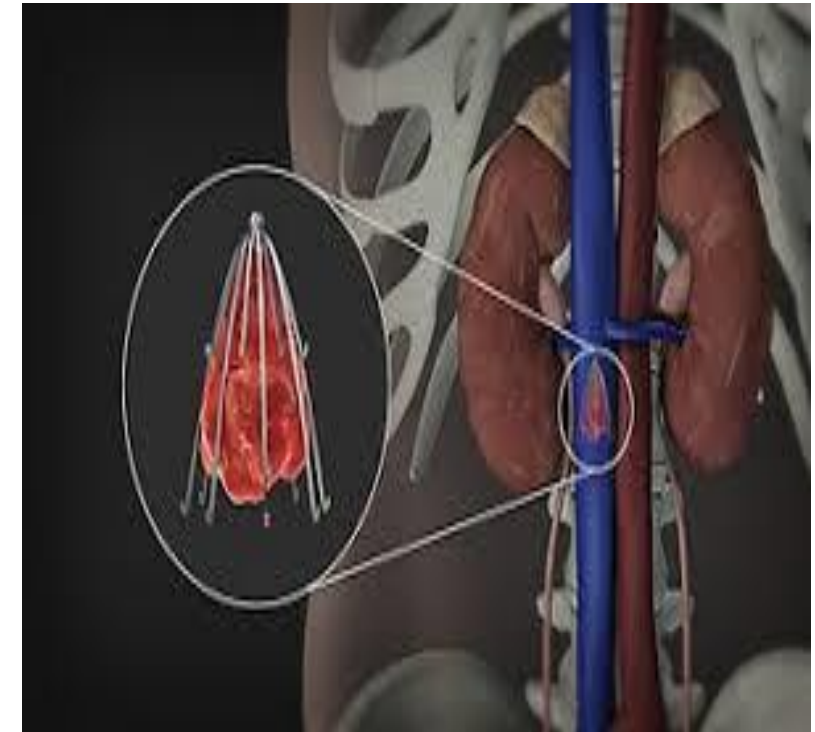
1. **Administration of unfractionated or LMWH and warfarin (Coumadin).**
 - Warfarin should be started at the same time and when the international normalized ratio (INR) reaches 3.0, the heparin should be discontinued.
 - The patient should remain on warfarin for 3 to 12 months, depending on his or her risk for thromboembolic disease.



Prevention of Recurrence

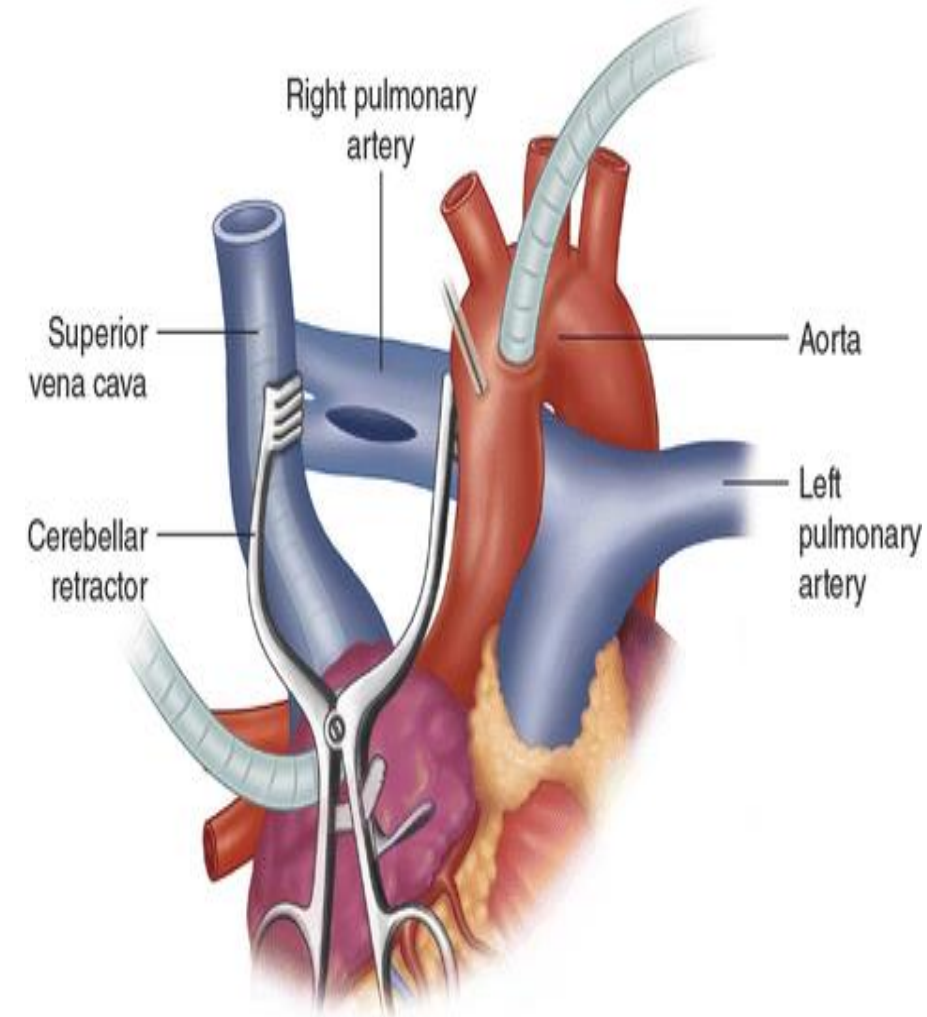
1. Heparin is administered to prevent further clots from forming and has no effect on the existing clot.
2. The INR should be maintained between 2.0 and 3.0 .
3. Interruption of the inferior vena cava is reserved for patients in whom anticoagulation is contraindicated.

The procedure involves **placement of a percutaneous venous filter (e.g., Green-field filter) into the vena cava, usually below the renal arteries**. The filter prevents further thrombotic emboli from migrating into the lungs



Clot Dissolution

1. The administration of fibrinolytic agents in the treatment of PE has had limited success.
2. Currently, fibrinolytic therapy is reserved for the patient with a massive PE and concomitant hemodynamic instability. Either **recombinant tissue –type plasminogen activator (rt-PA)** or **streptokinase** may be used.
3. If the fibrinolytic therapy is contraindicated, **pulmonary embolectomy** may be performed to remove clot.



Reversal of Pulmonary Hypertension

1. Administration of **inotropic agents** and fluid, fluids should be administered **to increase right ventricular preload, which would stretch the right ventricle and increase contractility,**
2. Thus overcoming the elevated pulmonary arterial pressure.
Inotropic agents also can be **used to increase contractility to facilitate an increase in CO.**

Nursing Management

Nursing actions are aimed

- ❖ preventing the development of DVT which is a major complication of immobility and a leading cause of PE. These **measures include**
 - Active/passive range-of-motion exercises involving foot extension
 - Use of pneumatic compression devices
 - Adequate hydration
 - Progressive ambulation.



Nursing interventions

- ✓ Optimizing oxygenation and ventilation,
- ✓ Monitoring for bleeding
- ✓ Providing comfort
- ✓ Emotional support
- ✓ Maintaining surveillance for complications
- ✓ Educating the patient and family.

Optimizing Oxygenation and Ventilation

- ❖ Positioning
- ❖ Preventing desaturation
- ❖ Promoting secretion clearance.

Monitoring for Bleeding

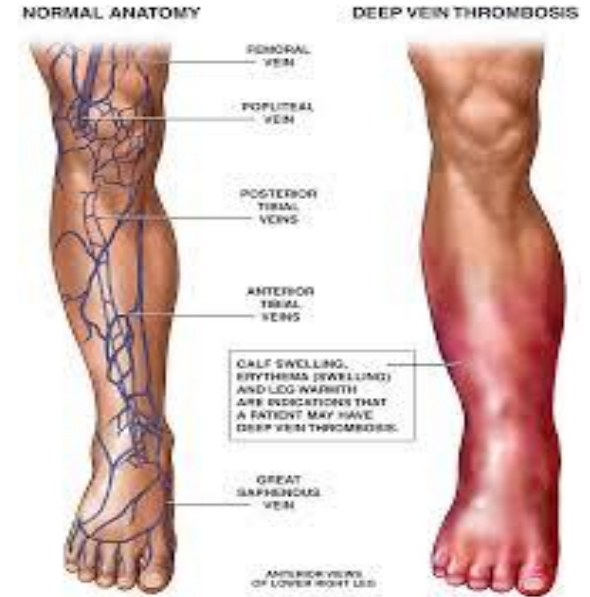
- The patient receiving anticoagulant or fibrinolytic therapy should be observed for signs of bleeding.
- The patient's gums, skin, urine, stool, and emesis should be screened for signs of overt or covert bleeding.
- In addition, monitoring the patient's INR or a PTT is critical to managing the anticoagulation therapy

Educating the Patient and Family

- ❑ Early in the patient's hospital stay, the patient and family should be taught about pulmonary embolus, its etiologies, and its treatment. As the patient moves toward discharge
- ❑ Teaching should focus on the interventions necessary for preventing the reoccurrence of DVT and subsequent emboli; signs and symptoms of DVT and anticoagulant complications and measures to prevent bleeding.
- ❑ If the patient smokes, he or she should be encouraged to stop smoking and be referred to smoking cessation program

Measures to prevent deep vein thrombosis (DVT)

- Avoid tight-fitting clothes, crossing legs, and prolonged sitting or standing, elevate legs when sitting, exercise.
- Signs and symptoms of DVT (e.g., redness, swelling, sharp or deep leg pain)
- importance of taking medications
- Signs and symptoms: of anticoagulant complications (e.g , excessive bruising: discoloration of the skin, changes in color of urine or stools)



Collaborative Management

1. Administer oxygen therapy .
2. Intubate patient .
3. Initiate mechanical ventilation.
4. Administer medications :
 - Fibrinolytic therapy Anticoagulants
 - Bronchodilators
 - Inotropic agents
 - Sedatives Analgesics

Collaborative Management

5- Administer fluids

6-Position patient to optimize ventilation/perfusion matching

8-Maintain surveillance for complications :

- Bleeding

- Acute respiratory distress syndrome

9- Provide comfort and emotional support



- **Which of the following is considered a non-thrombotic cause of pulmonary embolism?**
 - A. Deep vein thrombosis
 - B. Right ventricular clot
 - C. Fat embolism
 - D. Pelvic vein thrombus
- **Which of the following can lead to an air embolism causing PE?**
 - A. Central venous catheter placement
 - B. Long bone fracture
 - C. Tumor invasion
 - D. Atrial fibrillation
- **Which heart chamber can be a source of thrombotic emboli leading to PE?**
 - A. Left atrium
 - B. Left ventricle
 - C. Right atrium
 - D. Right ventricle



MANY THANKS

A graphic featuring the words "MANY THANKS" in a bold, white, 3D sans-serif font. The text is centered and appears to be floating above a series of vibrant, multi-colored brush strokes. These strokes, in shades of red, orange, yellow, green, blue, and purple, radiate outwards from behind the letters, creating a sense of dynamic energy and celebration. The entire composition is set against a plain white background.